Fractional-Charge Curvature and Electronic Gaps in Benzene

Gas-phase molecule: PBE/LibXC in SIESTA

Charles Kipchumba Rotich

28 September 2026

Three gap definitions for a molecule

$$E_g^{\mathrm{KS}} = \varepsilon_{\mathrm{LUMO}} - \varepsilon_{\mathrm{HOMO}},$$

$$E_g^{\mathrm{deriv}} = \left.\frac{\partial E}{\partial N}\right|_{30^+} - \left.\frac{\partial E}{\partial N}\right|_{30^-},$$

$$E_g^{\mathrm{integer}} = E(29) - 2E(30) + E(31).$$

The neutral pseudopotential calculation has 30 valence electrons.

Calculation setup

Molecule	Benzene, C_6H_6
Valence electrons	$6(4) + 6(1) = 30$
XC / pseudopotentials	PBE via LibXC / PBE PSML
Basis / sampling	DZP

Neutral benzene: Kohn–Sham HOMO–LUMO gap

Quantity	Energy (eV)
HOMO, band 15	-6.02302210
LUMO, band 16	-0.96614033

$$\boxed{E_g^{\mathrm{KS}} = 5.05688178\ \mathrm{eV}}$$

One Γ point; neutral, non-spin-polarized calculation.

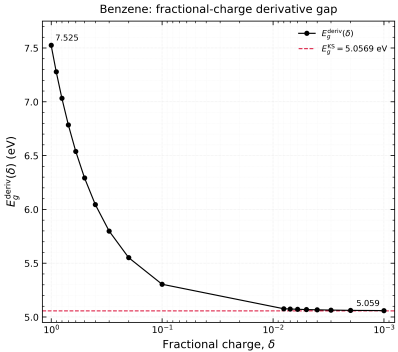
Integer ΔSCF at fixed geometry

State	Valence electrons	Total energy (eV)
Cation	29	−1069.059173
Neutral	30	−1076.392891
Anion	31	−1076.201803

$$I = 7.333718\ \mathrm{eV}, \qquad A = -0.191088\ \mathrm{eV}$$

$$\boxed{E_g^{\Delta\mathrm{SCF}} = I - A = 7.524806\ \mathrm{eV}}$$

Derivative gap: fractional charge



$$E_g(\delta) = \frac{E(30+\delta) - 2E(30) + E(30-\delta)}{\delta}.$$

At $\delta = 0.001$: 5.0590 eV.

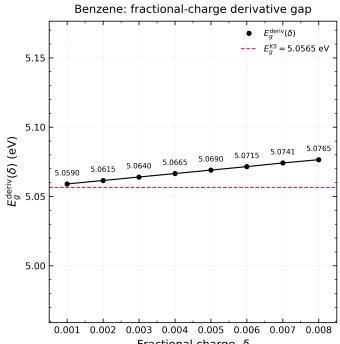
At $\delta = 0.1$: 5.30405 eV.

At $\delta = 1$: 7.524806 eV (integer ΔSCF).

As δ decreases, the gap tends toward the KS gap; convergence to within 0.0022 eV is achieved at $\delta = 0.001$.

These values describe different averaging intervals, not a single charge-independent gap.

Small-step derivative gap approaches the KS gap



$$E_g(\delta) = \frac{E(30+\delta) - 2E(30) + E(30-\delta)}{\delta}.$$

δ	$E_g(\delta)$ (eV)
0.001	5.059000
0.002	5.061500
0.004	5.066500
0.008	5.076500

KS reference: 5.056882

eV. At $\delta = 0.001$, difference $= 0.002118$ eV.

Current results and next calculations

Method	Gap (eV)
KS HOMO–LUMO	5.056882
Derivative, $\delta = 0.001$	5.059000
Integer ΔSCF	7.524806